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Efficient Synthesis of an Unprecedented Enantiopure Hybrid Scorpionate/Cyclopentadienyl by Diastereoselective Nucleophilic Addition to a Fulvene

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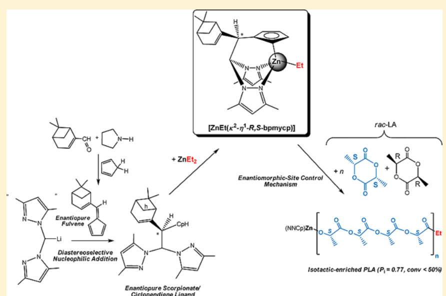
Supporting Information

ABSTRACT: The work described here represents the first example of an enantiopure hybrid scorpionate/cyclopentadiene ligand. The ligand was obtained in a one-pot synthetic procedure by an efficient and highly diastereoselective nucleophilic addition of an organolithium reagent to an electrophilically activated olefin in a new fulvene with a chiral substrate to control the stereochemistry of a newly created asymmetric center. We verified the potential utility of this ligand as a valuable scaffold that is able to induce chirality in organometallic/coordination chemistry. This was achieved through the preparation of a new enantiomerically pure zinc complex in which the ligand behaves in a tridentate manner with a κ^2NN-$\eta^1(\pi)$-Cp coordination mode. This alkylzinc complex constitutes the first example of an organozinc derivative which behaves as an efficient initiator for the ROP of *rac*-LA in the production of isotactic-enriched poly(lactides) with P_i = 0.77.



Research on metal catalysts bearing stereogenic centers has increased in recent years due to the current interest in asymmetric catalysis for the preparation of fine chemicals.[1] One important task in this field is the design of enantiomerically rich multifunctional ligands that are able to induce chirality on the metal center.[2] In recent years, one of our research areas has concerned the preparation of multifunctional ligands based on bis(pyrazol-1-yl)methane building blocks.[3] For example, we previously reported the synthesis and structural characterization of the first lithium salts of several hybrid scorpionate/cyclopentadienyl ligands by a simple one-pot procedure (addition of an organolithium to a fulvene).[4] With this in mind, we focused our attention on enantiopure fulvene compounds[5] as possible chiral substrates to control the stereochemistry of a newly created asymmetric center. Conjugate addition of organometallic reagents to electrophilically activated olefins is a versatile methodology to form a new carbon−carbon bond.[6] Regioselectivity for 1,4-additions or Michael reactions of organolithium reagents with activated olefins can be controlled by a chiral auxiliary or an external chiral ligand, and these processes are known for α,β-unsaturated esters, imines, and other electron-withdrawing groups.[7] However, only two examples of enantioselective addition to a fulvene (where the electron-withdrawing group is the cyclopentadiene ring) have been reported,[5,8] generating a chiral cyclopentadienide anion. The cyclopentadienyllithium was generated by the asymmetric addition of methyllithium or aryllithium to a fulvene substituted with a chiral amino group[5] or 6-(dimethylamino)fulvene in the presence of the chiral ligand (−)-sparteine,[8] respectively. These cyclopentadienyl synthons (not isolated) were used for the asymmetric synthesis of metallocenes. In the present work we developed a simple and efficient synthetic route that allowed us to isolate the first enantiopure hybrid scorpionate/cyclopentadiene compound as a precursor for a new class of tridentate ligands. This route constitutes an unusual example of a highly enantioselective addition to a new enantiopure fulvene. This fulvene was synthesized by a simple one-pot procedure, with a chiral substrate to control the stereochemistry of a newly created asymmetric center to give the scorpionate/cyclopentadiene compound, which was isolated and transferred onto Zn(II) to give an enantiopure scorpionate−zinc complex. This complex was assessed as a single-component initiator for the ring-opening polymerization (ROP) of lactides, producing an isotactic enrichment in the resulting poly(lactides) (PLAs).

As discussed above, optically active 6-(dialkylamino)fulvenes have previously been used to prepare chiral cyclopentadienyl synthons,[5] but these fulvenes were obtained in 10−64% overall yield in a four- to five-step procedure. Therefore, the first step of our synthetic strategy was to design a one-pot procedure to synthesize a new enantiopure fulvene in good yield. It was found that the reaction of the commercially available (1*R*)-(−)-myrtenal and cyclopentadiene in the presence of

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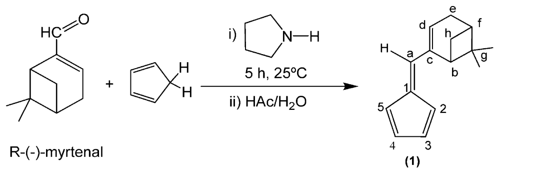
pyrrolidine gave the new enantiopure fulvene 2-(2,4-cyclopentadienylidenemethyl)-(1*R*)-6,6-dimethylbicyclo[3.1.1]-2-heptene (**1**), which was obtained as a red oil in good yield (ca. 90%) (Scheme 1). The ^{1}H NMR spectrum of **1** (Figure S1 in

Scheme 1. Synthesis of Enantiopure Fulvene 1

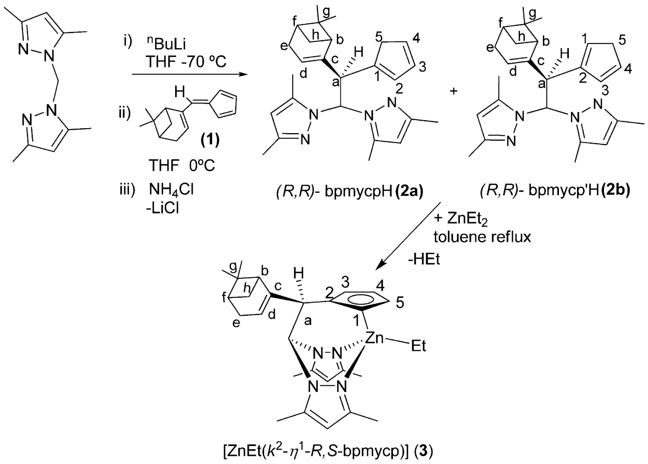


the Supporting Information) shows the corresponding four signals (H^2, H^3, H^4, and H^5) for the asymmetric cyclopentadiene of fulvene. In addition, the spectrum exhibits five sets of resonances for H^b, H^d, H^e, H^f, and *gem*-H^h and three singlets for H^a and the methyl groups of the bicyclic substituent on the carbon atom. ^{1}H−^{13}C heteronuclear correlation (g-HSQC) experiments allowed us to assign the resonances corresponding to the ^{13}C{^{1}H} NMR spectrum of this compound.

Once the new optically active fulvene had been synthesized, we focused our attention on the second step, i.e., the preparation of enantiopure scorpionate ligands, using the aforementioned methodology based on the asymmetric nucleophilic addition of organolithium reagents to C═C bonds. Thus, a cold (0 °C) THF solution of lithium bis(3,5-dimethylpyrazol-1-yl)methide,[9] prepared in situ from ^{n}BuLi and bis(3,5-dimethylpyrazol-1-yl)methane (bdmpzm)[10] at −70 °C, was added dropwise to a THF solution containing 1 equiv of the new fulvene **1**. The addition gave rise to a rapid color change from yellow to colorless. The reaction was complete after 5 min, and the appropriate workup gave a 1/1 mixture of the two scorpionate/cyclopentadiene regioisomers, both enantiopure compounds, (*R*,*R*)-bpmycpH (**2a**) (bpmycpH = 1-{2,2-bis(3,5-dimethyl-1*H*-pyrazol-1-yl)-1-(2*R*)-2-[(1*R*)-6,6-[3.1.1]-2-hepten-2-yl]ethyl}cyclopentadiene) and (*R*,*R*)-bpmycp′H (**2b**) (bpmycp′H = 2-{2,2-bis(3,5-dimethyl-1*H*-pyrazol-1-yl)-1-(2*R*)-2-[(1*R*)-6,6-[3.1.1]-2-hepten-2-yl]ethyl}-cyclopentadiene) as colorless solids, in 63% yield, and with an excellent diastereomeric excess (>99% de) (Scheme 2). This procedure constitutes an efficient and highly diastereoselective method to prepare enantiopure scorpionate ligands in a one-pot process.[11] Initial evidence for the stereochemical route was obtained by X-ray diffraction (see below), which shows that the newly formed chiral center has the *R* configuration.

This diastereoselectivity was assessed by considering the ^{1}H NMR spectrum and integrating the CH^a or Me signals of the bicycle, in the mixture of regioisomers, since the chemical shifts of these protons appear systematically at higher field for the *S* epimer. A diastereomeric ratio denoted as ">99:1" signifies that only the major diastereoisomer was detected. The ^{1}H NMR spectrum of **2** (Figure S2 in the Supporting Information) exhibits four singlets for each of the H^4, Me^3, and Me^5 pyrazole protons, indicating the presence of two tautomers, which indicates that the two pyrazole rings are inequivalent. The spectrum also contains two doublets—one for the CH bridge to two pyrazole rings and one for CH^a. One set of signals for the cyclopentadiene group and the bicycle was observed for each tautomer. In both regioisomers (**2a**,**b**) the H^5 signal in the ^{1}H NMR spectrum integrates for two protons, meaning that

Scheme 2. Synthesis of the Enantiopure Heteroscorpionate Ligands (*R*,*R*)-bpmycpH (2a) and (*R*,*R*)-bpmycp′H (2b) and the Zinc Complex [ZnEt(κ^2-η^1-*R*,*S*-bpmycp)] (3)



regioisomer **2c** (in which the cyclopentadiene group is bonded by C^5 to the bis(pyrazol-1-yl)methane unit) can be ruled out.

Having prepared this new enantiopure heteroscorpionate ligand in the form of the cyclopentadiene, we explored its potential utility as a tridentate ligand in the preparation of enantiopure group 12 metal complexes. Deprotonation in the cyclopentadiene group of **2** with $ZnEt_2$ in a 1/1 molar ratio yielded the enantiopure zinc complex [ZnEt(κ^2-η^1-*R*,*S*-bpmycp)] (**3**; *R*,*S*-bpmycp = 2-{2,2-bis(3,5-dimethyl-1*H*-pyrazol-1-yl)-1-(2*S*)-2-[(1*R*)-6,6-[3.1.1]-2-hepten-2-yl]ethyl}-cyclopentadienyl), which was isolated as a colorless solid in excellent yield (90%) (Scheme 2). It should be noted that the coordination of the cyclopentadienyl moiety to the metal in this complex produces a change in the configuration descriptor of the carbon C^a (*S*) with respect to that in compound **2** (*R*) but the spatial arrangement around this carbon atom is the same in both compounds. Complex **3** constitutes the first example of a metal complex bearing an enantiopure hybrid scorpionate/cyclopentadienyl ligand.

The ^{1}H NMR spectrum of complex **3** (Figure S3 in the Supporting Information) shows two singlets for each of the H^4, Me^3, and Me^5 pyrazole protons, one doublet for each of the methine groups (CH bridge of pyrazole rings and CH^a), four multiplets for protons H^b, H^d, H^e, and H^f, and two multiplets for the *gem*-H^h bicycle protons. The ^{1}H NOESY-1D experiments allowed the unequivocal assignment of all ^{1}H resonances. Thus, H^1 of the cyclopentadienyl group (δ 7.09 ppm) shows significant deshielding with respect to the equivalent proton in complexes with η^5-Cp coordination[12] and this is consistent with a η^1 coordination through this carbon atom (C^1), indicating an identical molecular structure in solution and the solid state.

As mentioned above, the absolute configurations of **2** and **3** were verified by single-crystal X-ray diffraction analysis of complex **3** (Figure 1).[13] The zinc atom has a distorted-tetrahedral geometry in which the heteroscorpionate ligand acts in a tridentate fashion and is coordinated by the two nitrogens of the pyrazole rings and one carbon of the cyclopentadienyl group, with the ethyl ligand occupying the fourth site of the tetrahedron. The most notable feature in the X-ray molecular structure of complex **3** is the peripheral $\eta^1(\pi)$ coordination mode of the cyclopentadienyl ring. The differences (Δd) between the $C_{\alpha\beta}$ and $C_{\beta\beta}$ bonds (0.005 and 0.02 Å) and the

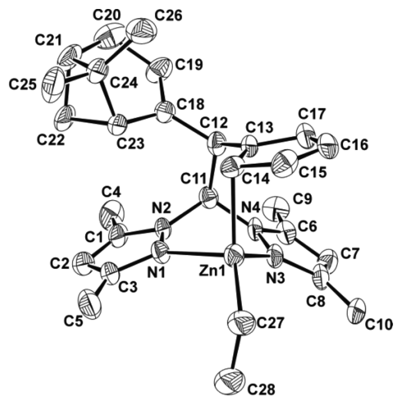


Figure 1. ORTEP diagram of **3** with 30% probability ellipsoids. Selected bond lengths (Å) and angles (deg): Zn(1)−N(1), 1.372(7); Zn(1)−N(3), 2.086(6); Zn(1)−C(14), 2.137(6); Zn(1)−C(27), 1.973(6); N(3)−Zn(1)−N(1), 86.0(2); N(3)−Zn(1)−C(14), 102.3(2); C(27)−Zn(1)−C(14), 128.4(3); N(1)−Zn(1)−C(14), 94.2(2); C(27)−Zn(1)−N(3), 119.6(3); C(27)−Zn(1)−N(1), 116.0(3).

distances between C_α and the zinc-bound carbon atom C(14) are somewhat smaller (C(13)−C(14) = 1.431(8) Å; C(14)−C(15) = 1.44(1) Å) than expected for a single C−C bond that has significant π-electron delocalization. These data, as well as the angle between the Zn−C bond and the plane of the cyclopentadienyl ring (102.6°), are in agreement with similar results for zinc complexes synthesized by our research group[12] and also for zincocenes in which one of the two cyclopentadienyl moieties is in a similar $\eta^1(\pi)$ coordination mode.[14] Additionally, the Zn(1)−C(14) bond length of 2.137(6) Å is consistent with that previously observed in zincocenes with parallel ($\eta^5/\eta^1(\pi)$) and nonparallel ($\eta^5/\eta^1(\sigma)$) cyclopentadienyl rings.[14] The distances between Zn(1) and the carbon atoms C(13) (2.788(3) Å), C(15) (2.734(8) Å), C(16) (3.407(8) Å), and C(17) (3.445(7) Å) can be considered as nonbonding.[12,14]

Finally, zinc-based[12,15] catalysts have been extensively employed in ROP for the production of PLAs due to their biocompatibility[16] and efficiency as initiators[17] in a wide array of applications (packaging, microelectronics, and biomedical).[18] As a result, we decided to carry out a preliminary assessment of the catalytic activity of this zinc alkyl (**3**) in the ROP process under different conditions. Inspection of the experimental M_n values of the PLAs produced revealed that the molecular weights are very close to the expected theoretical values ($M_{n(calcd)}(PLA_{100})$ = 14400) (Table 1). Size exclusion chromatography (SEC) data for the resulting polyesters also showed a monomodal weight distribution under mild conditions, with polydispersities ranging from 1.08 to 1.13 (Figure S4 in the Supporting Information). These results are characteristic of well-controlled living propagations and the existence of a single type of reaction site. Thus, **3** acted as an active single-component initiator, and 62% of L-LA was transformed at 50 °C after 1.5 h (Table 1, entry 2). The use of tetrahydrofuran as solvent led to a dramatic decrease in the catalytic activity (Table 1, entry 3). MALDI-ToF MS (Figure S5 in the Supporting Information) of low-molecular-weight materials provided evidence that the ring opening of L-LA occurs by the initial addition of the alkyl fragment to the monomer. Initiator **3** was also tested in the polymerization of *rac*-lactide, and not surprisingly, this proved to be slower than for L-LA. For instance, 47% of the monomer was converted after 4 h at 50 °C, with low-molecular-weight PLA materials and narrow polydispersity values obtained (Table 1, entry 5; M_n = 6600, M_w/M_n = 1.13). Microstructural analysis of the poly(*rac*-lactide) revealed that this initiator exerts a significant preference for isotactic dyad enchainment, reaching P_i = 0.77 (Table 1 and Figure S6), a remarkable result considering that this is the first zinc-containing example reported to date that is capable of achieving this value, since none of the very scarce zinc catalysts bearing chiral auxiliaries[19] have succeeded in the isoselective ROP of *rac*-LA.[20] Additionally, analysis of the tetrads resulting from stereoerrors (i.e., tetrads other than *iii*) indicated that an enantiomorphic site control mechanism[21] is dominant ([*sis*]/[*sii*]/[*iis*]/[*isi*] = 1/1/1/2; Figure S6 and Table 1). This behavior for catalyst **3** is most probably the result of the high homosteric control caused by the enantiopure hybrid scorpionate/cyclopentadienyl ligand synthesized.

In conclusion, we present here a one-pot synthetic procedure to obtain the first enantiopure hybrid scorpionate/cyclopentadiene ligand through the highly diastereoselective nucleophilic addition of a lithium bis(3,5-dimethylpyrazol-1-yl)methide derivative to a new enantiopure fulvene, which bears a chiral substrate to control the stereochemistry of the newly created asymmetric center. In addition, this compound proved to be an excellent reagent to introduce chirality in metal complexes—a fact confirmed by reaction with diethylzinc. This alkylzinc represents the first such complex that is capable of producing isotactic-enriched poly(lactides) (P_i = 0.77) from *rac*-LA.

Table 1. Polymerization of L-/*rac*-Lactide Catalyzed by 3[a]

entry	monomer	temp (°C)	time (h)	yield (g)	conversn (%)[b]	$M_{n(theor)}$ (Da)[c]	M_n (Da)[d]	M_w/M_n[d]	P_i[e]
1	L-LA	20	12	0.74	57	8200	8400	1.10	
2	L-LA	50	1.5	0.80	62	8900	8700	1.12	
3	L-LA[f]	50	3	0.32	25	3600	3400	1.13	
4	*rac*-LA	20	15	0.41	32	4600	4800	1.08	0.77
5	*rac*-LA	50	4	0.61	47	6800	6600	1.13	0.73

[a]Polymerization conditions: $[\mathbf{3}]_0$ = 90 μmol of catalyst, 25 mL of toluene as solvent, $[\text{L-}/rac\text{-lactide}]_0/[\mathbf{3}]_0$ = 100. [b]Percentage conversion of the monomer ((weight of polymer recovered)/(weight of monomer) × 100). [c]Theoretical M_n = (monomer/catalyst) × (% conversion) × (M_w of lactide). [d]Determined by size exclusion chromatography relative to polystyrene standards in tetrahydrofuran. The experimental M_n was calculated considering Mark−Houwink's corrections[22] for M_n (M_n(obsd) = 0.56 × M_n(GPC)). [e]The parameter P_i (i = isotactic) is the probability of forming a new *i* dyad. The P_i and the P_s (s = syndiotactic) values were calculated from the following tetrad probabilities based on enantiomophic site control statistics[21c] in the polymerization of *rac*-lactide: *sis, sii, iis*, $[P_i^2(1-P_i) + P_i(1-P_i)^2]/2$; *iii*, $[P_i^2(1-P_i)^2 + P_i^3 + (1-P_i)^3]/2$; *isi*, $[P_i(1-P_i) + P_i(1-P_i)]/2$. [f]10 mL of tetrahydrofuran as solvent.

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■ ASSOCIATED CONTENT

Supporting Information

Figures, tables, text, and a CIF files giving details of the synthesis and spectroscopic data for all compounds details of data collection, refinement, atom coordinates, anisotropic displacement parameters, and bond lengths and angles for **3**, and details of the ring-opening polymerization of lactides with complex **3**. This material is available free of charge via the Internet at http://pubs.acs.org.

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Notes

The authors declare no competing financial interest.

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(13) Crystallographic data for **3**: $C_{28}H_{38}N_4Zn$, orthorhombic, $P2_12_12_1$; a = 9.892(3) Å, b = 15.484(5) Å, c = 17.079(6) Å; V = 2616(2) Å^3; Z = 4; D_{calcd} = 1.259 g cm^{-3}; λ(Mo Kα) = 0.71073 Å; μ(Mo Kα) = 0.961 mm^{-1}; 230(2) K; X8 APEX II CCD diffractometer, graphite monochromator, 4699 unique reflections; R = 0.0623, R_w = 0.1257.

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